

1 Type-I calibration of fixed-effect interaction tests in
2 linear mixed models: a staged diagnosis of
3 working-covariance mis-specification and a
4 cluster-robust remedy

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32 Abstract

33 **Background.** The linear mixed model is the standard tool for testing a fixed-
 34 effect interaction in longitudinal and clustered data. The model-based Wald test
 35 of that interaction is exact only when the working within-cluster covariance is
 36 correctly specified. In practice an analyst chooses a convenient parametric residual
 37 structure, a continuous-time autoregressive correlation, compound symmetry, or
 38 an unstructured form, and then trusts the model-based standard error. The
 39 direction and the magnitude of the mis-calibration that follows when the working
 40 covariance is wrong are rarely checked.

41 **Methods.** We set out a staged diagnostic protocol, a four-channel decomposition
 42 of the test statistic, a working-covariance model ladder, and a sample-size gradient,
 43 and apply it to a worked example: the test of a biomarker-by-treatment interaction
 44 in an aggregated N-of-1 carryover simulation, analysed with an `lme` model carrying
 45 a random intercept and a `corCAR1` residual correlation.

46 **Results.** At the null the model-based standard error of the interaction is over-
 47 estimated by 6 to 10 percent, and the realised type-I error is approximately 0.03
 48 against a nominal 0.05. The degrees-of-freedom rule, the positive-definiteness
 49 correction, and point-estimate bias are excluded as causes. The model ladder
 50 localises the mis-calibration to the `corCAR1` residual layer: a random intercept
 51 alone is well calibrated. The sample-size gradient shows the bias is asymptotic,
 52 not a moderate-sample artefact: it does not attenuate from $N = 35$ to $N =$
 53 280 . A CR2 bias-reduced cluster-robust standard error restores the standard-
 54 error ratio to approximately one and the type-I error to approximately 0.05. At
 55 an alternative cell with a non-null effect the recalibration raises power by a near-
 56 uniform increment and leaves the ranking of analysis specifications unchanged.

57 **Conclusions.** A mis-specified parametric working covariance can bias the model-
 58 based standard error of a mixed-model interaction test asymptotically, in either
 59 direction. Cluster-robust standard errors are a general remedy, and the staged
 60 diagnostic is a reusable instrument for detecting the problem.

61 1 Introduction

62 **The Wald test of a mixed-model interaction is exact only when the work-**
 63 **ing covariance is correctly specified.**

- 64 • The estimand is a fixed-effect interaction coefficient.
- 65 • Inference proceeds by dividing the estimate by a model-based standard
- 66 error and referring to a t distribution.
- 67 • Exactness requires the assumed random-effects structure and residual cor-
- 68 relation to match the truth.

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76 *The linear mixed model is, by a wide margin, the most common vehicle*
77 *for testing whether an effect is moderated by a covariate in longitudinal or*
78 *clustered data [@laird1982; @diggle2002]. The quantity of interest is a fixed-*
79 *effect interaction coefficient, and the inference almost always proceeds through*
80 *a Wald test: the coefficient estimate is divided by a model-based standard*
81 *error and referred to a t distribution. That procedure is exact only under a*
82 *condition that is easy to state and, unfortunately, easy to forget. The working*
83 *covariance, that is, the random-effects structure together with the parametric*
84 *residual correlation the analyst has assumed, must be correct.*

85 **Covariance mis-specification renders the model-based standard error in-**
86 **consistent, motivating the cluster-robust sandwich estimator.**

- 87 • Eicker, Huber, and White established that a mis-specified working covari-
88 ance yields an inconsistent model-based standard error.
- 89 • Liang and Zeger introduced GEE, decoupling point estimation from vari-
90 ance estimation via an empirical sandwich.
- 91 • A substantial literature has since refined the sandwich estimator for the
92 small-sample regime.

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100 *It has been understood since the work of Eicker, Huber, and White that*
101 *this condition is more demanding than it appears [@eicker1967; @huber1967;*
102 *@white1980]. When the working covariance is mis-specified, the model-based*
103 *standard error is no longer a consistent estimator of the true sampling vari-*
104 *ability of the fixed-effect estimate. Liang and Zeger [-@liang1986] carried this*
105 *insight into longitudinal data analysis: their generalized estimating equations*
106 *decouple the point estimate, computed under a convenient working correla-*
107 *tion, from the variance estimate, computed by an empirical sandwich that*
108 *remains consistent whatever the working correlation happens to be. The*
109 *sandwich, or cluster-robust, variance estimator has since become a standard*
110 *instrument, and a substantial literature has refined it for the small-sample*
111 *case [@mancl2001; @bell2002; @cameron2015; @pustejovsky2018].*

112 **The direction and persistence of mis-calibration from a plausible but in-**
113 **correct working covariance are rarely audited and cannot be answered in**
114 **the abstract.**

- 115 • Analysts routinely adopt parametric residual correlations on substantive

- 116 grounds and trust the resulting standard errors.
- 117 • The magnitude, direction, and sample-size behaviour of the resulting mis-
- 118 calibration depend on the specific interaction and covariance mismatch.
- 119 • Answering these questions requires examining a concrete analysis.

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127 *Yet in routine mixed-model practice the working covariance is still trusted.*

128 *An analyst fitting longitudinal data with `nlme` or a comparable package will*

129 *choose a residual correlation, very often a continuous-time first-order au-*

130 *toregressive form, because it is pharmacokinetically or biologically plausible*

131 *[@pinheiro2000], and will then report the model-based standard error of the*

132 *interaction without asking what happens if the chosen form is not exactly*

133 *the form the data follow. This is not negligence so much as a reasonable*

134 *working assumption that is seldom audited. But how large is the error when*

135 *the assumption fails? Is the resulting test conservative or anti-conservative?*

136 *Does the mis-calibration diminish as the sample grows, so that it can perhaps*

137 *be dismissed as a small-sample nuisance, or does it persist? These questions*

138 *do not have a single answer, because the answer depends on the interaction*

139 *being tested and on the way the working covariance departs from the truth.*

140 *They have to be asked of a particular analysis, and we shall ask them of one.*

141 **This paper contributes a reusable four-channel diagnostic protocol and a**

142 **worked finding of asymptotic, conservative mis-calibration in a longitudi-**

143 **nal interaction test.**

- 144 • The protocol decomposes the Wald statistic into bias, scale, degrees of
- 145 freedom, and reference-shape channels, supplemented by a model ladder
- 146 and a sample-size gradient.
- 147 • The worked example reveals a conservative test whose standard error is
- 148 over-estimated by roughly ten percent.
- 149 • The conservatism is asymptotic and is removed by a cluster-robust standard
- 150 error.

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158 *In this paper we endeavour to do exactly that for one analysis, and in doing*

159 *so we offer two contributions that we believe are more general than the single*

160 *case. The first is a diagnostic protocol. We decompose the Wald statistic*

161 *into four channels through which mis-calibration can enter, and we add two*

further instruments: a working-covariance model ladder that localises the cause, and a sample-size gradient that separates a finite-sample artefact from an asymptotic one. The protocol is not specific to the example and can be applied to any mixed-model interaction test whose calibration is in doubt. The second contribution is the finding the protocol produced. In our worked example the model-based interaction test is conservative, not anti-conservative as the sandwich literature's emphasis on under-estimated standard errors might lead one to expect; the conservatism is material, on the order of ten percent in standard-error terms; and it is asymptotic, persisting undiminished across an eightfold range of sample size. We trace it to a single component of the working covariance, and we show that a cluster-robust standard error removes it.

The worked example is drawn from a carryover simulation study and serves solely to make the general phenomenon concrete.

- The example is the biomarker-treatment interaction test in the pmsimstats-ng carryover-sensitivity study.
- The finding is a property of the mixed-model Wald test, not of the particular simulation.
- Sections 2 through 5 address the protocol, the example, the staged results, and their generalisation.

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The worked example is drawn from the pmsimstats-ng project, a simulation study of power to detect biomarker-treatment interactions in aggregated N-of-1 trials under carryover [hendrickson2020]. The example is incidental to the argument. The phenomenon, a parametric working covariance that biases a fixed-effect interaction test asymptotically, and the remedy are not properties of that simulation; they are properties of the mixed-model Wald test, and the example serves only to make them concrete and quantified. We begin in Section 2 with the diagnostic protocol. Section 3 describes the worked example. Section 4 reports the staged diagnosis and the cluster-robust remedy. Section 5 discusses the generality of the result and its limits.

2 A staged diagnostic protocol

Mis-calibration of a Wald test must enter through exactly one of four separable channels, each diagnosable by simulation.

- The test statistic is $T = \hat{\theta}/\widehat{SE}$, referred to a t_ν critical value.
- Correct calibration requires the realised rejection rate to equal α under H_0 .
- The four channels – bias, scale, degrees of freedom, and reference shape – are mutually exhaustive and separable.

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213 *Consider a fixed-effect interaction coefficient θ estimated by $\hat{\theta}$, with a model-*
 214 *based standard error $\widehat{\text{SE}}$ and an assigned degrees of freedom ν . The Wald test*
 215 *rejects the null $H_0 : \theta = 0$ at level α when the statistic $T = \hat{\theta}/\widehat{\text{SE}}$ exceeds*
 216 *the critical value $t_{\nu, 1-\alpha/2}$ in absolute value. The test is correctly calibrated*
 217 *when the realised rejection rate under H_0 equals α . When it does not, the*
 218 *discrepancy must enter through one of exactly four channels, and the channels*
 219 *are separable by simulation. We begin with the decomposition itself, and then*
 220 *describe the two further instruments we shall add to it.*

221 2.1 The four-channel decomposition

222 We begin by writing the interaction estimate in a form that exposes the four
 223 channels. Let

$$\hat{\theta} = \mu + \sigma Z^*, \quad \mu = E(\hat{\theta}), \quad \sigma = \text{SD}(\hat{\theta}),$$

224 so that $Z^* = (\hat{\theta} - \mu)/\sigma$ is the standardised estimate, of mean zero and unit variance
 225 by construction; when $\hat{\theta}$ is approximately normal, Z^* is approximately standard
 226 normal. Write $\widehat{\text{SE}}$ for the mean of the model-based standard error and $r = \widehat{\text{SE}}/\sigma$
 227 for its ratio to the true sampling standard deviation σ . Then the Wald statistic,
 228 to the order at which the variability of $\widehat{\text{SE}}$ about its mean may be neglected, is

$$T = \frac{\hat{\theta}}{\widehat{\text{SE}}} \approx \frac{\mu + \sigma Z^*}{r \sigma} = \frac{1}{r} (Z^* + \delta), \quad \delta = \frac{\mu}{\sigma},$$

229 where δ is the bias expressed in standard-deviation units. The realised type-I
 230 error follows at once,

$$\alpha_{\text{real}} = \Pr(|T| > t_{\nu, 1-\alpha/2}) \approx \Pr(|Z^* + \delta| > r t_{\nu, 1-\alpha/2}),$$

231 and exactly four quantities appear in it. They are the four channels: the stan-
 232 dardised bias δ , which decentres the statistic; the scale r , which multiplies the
 233 rejection threshold; the degrees of freedom ν , which fix the critical value $t_{\nu, 1-\alpha/2}$;
 234 and the distribution of Z^* , the shape of the reference. When all four take their
 235 nominal values, $\delta = 0$, $r = 1$, $\nu = \infty$, and $Z^* \sim N(0, 1)$, the expression collapses
 236 to $\Pr(|N(0, 1)| > z_{1-\alpha/2}) = \alpha$ and the test is exact; a departure in any one channel
 237 mis-calibrates it. A second identity is worth recording, since the diagnosis turns
 238 on it: because $T \approx (Z^* + \delta)/r$, the dispersion of the statistic satisfies

$$\text{SD}(T) \approx \text{SD}(Z^*)/r,$$

so a statistic standard deviation below one is read at once as a scale inflation ($r > 1$), a sub-normal reference shape, or both.

We measure the four channels by simulation as follows.

1. **Point-estimate bias** (δ). If $E(\hat{\theta}) \neq 0$ under H_0 the statistic is decentred. We measure μ , and hence $\delta = \mu/\sigma$, by the mean of $\hat{\theta}$ across null replicates.
2. **Standard-error calibration** (r). If the model-based $\widehat{SE}$ does not estimate the true sampling standard deviation of $\hat{\theta}$, the statistic is rescaled. We measure r as the mean model-reported standard error over the empirical standard deviation of the estimate across replicates. A ratio above one indicates an over-stated standard error and a conservative test; a ratio below one the reverse.
3. **Degrees of freedom** (ν). If ν is too small the t reference has tails too heavy and the test is conservative; if too large, the reverse. We measure this by recomputing the p-values against a normal reference and comparing the realised type-I error.
4. **Reference-distribution shape** (the law of Z^*). Even with $\delta = 0$, $r = 1$, and ν correct, Z^* may not be standard normal. We measure this by the standard deviation of T under H_0 , which should be approximately one, and by the empirical distribution function of the p-values, which should be $\text{Uniform}(0, 1)$.

The decomposition is novel not for introducing new measures but for showing that exactly these four quantities compose into the realised type-I error, making the diagnosis exhaustive and localising.

- Each channel – bias, scale, degrees of freedom, shape – corresponds to an established performance measure.
- Together they are exhaustive: a discrepancy absent from all four cannot exist.
- Because the channels enter the expression in distinct, non-interacting roles, the decomposition localises the fault rather than merely detecting it.

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The individual channels are not new. The bias of the estimate and the ratio of the model-based to the empirical standard error are among the standard performance measures of a simulation study, as codified by Morris, White, and Crowther [-@morris2019simulation], who recommend that the model-based and empirical standard errors be reported and compared as a matter of course; the degrees of freedom of the reference t are the subject of the small-sample literature of Satterthwaite [-@satterthwaite1946] and Kenward and Roger [-@kenward1997]; and the uniformity of the null p-value distribution is the

283 elementary calibration check that follows from the probability integral trans-
 284 form. What the decomposition contributes is not a new measure but the
 285 observation that these four quantities, and exactly these four, compose into
 286 the single expression for α_{real} derived above. Two consequences follow. First,
 287 the channels are exhaustive: a Wald test is a centred statistic, a scale, a
 288 reference shape, and a degrees-of-freedom choice, and the realised error rate
 289 is a function of those and of nothing else, so a discrepancy absent from all
 290 four cannot exist. Second, because the channels enter the expression in dis-
 291 tinct and non-interacting roles, measuring all four does not merely detect a
 292 mis-calibration but localises it, identifying which term carries the fault. Since
 293 the channels have different causes and different remedies, a diagnosis is in
 294 this sense a decomposition and not a checklist.

295 2.2 The working-covariance model ladder

296 The four-channel decomposition tells us how a test is mis-calibrated. It does
 297 not tell us why. When the decomposition points to the standard-error channel,
 298 it is worth being precise about what the scale r measures. Let the interaction
 299 estimate be written $\hat{\theta} = \ell' \hat{\beta}$, with ℓ the contrast vector that selects the interaction
 300 coefficient from the fixed-effects vector. A fixed-effect estimator that solves a
 301 generalized least squares system under a working covariance V_0 , when the data
 302 are in truth generated under a covariance V , has a model-based variance

$$v_0 = \ell'(X'V_0^{-1}X)^{-1}\ell,$$

303 while its true sampling variance is the sandwich form

$$v = \ell'(X'V_0^{-1}X)^{-1}(X'V_0^{-1}VV_0^{-1}X)(X'V_0^{-1}X)^{-1}\ell,$$

304 where X is the fixed-effects design and, the clusters being independent, every
 305 cross-product is a sum over clusters, $X'V_0^{-1}X = \sum_i X_i'V_{0i}^{-1}X_i$ and likewise
 306 $X'V_0^{-1}VV_0^{-1}X = \sum_i X_i'V_{0i}^{-1}V_iV_{0i}^{-1}X_i$ ^{1,2}. The scale of Section 2.1 is the ratio
 307 of these two quadratic forms, $r^2 = v_0/v$. They coincide when $V_0 = V$, and for
 308 a generic contrast ℓ they do not otherwise; the scale channel is therefore not
 309 a finite-sample nuisance but a direct consequence of the working covariance:
 310 r departs from one whenever $V_0 \neq V$, and no quantity of data removes the
 311 discrepancy.

312 **A model ladder localises the covariance component responsible for**
 313 **standard-error mis-calibration by tracking r across a graded sequence of**
 314 **working covariances fitted to the same data.**

- 315 • Each rung substitutes a different working covariance V_0 for the identical
 316 simulated data.
- 317 • The rung at which r departs from one identifies the responsible component.
- 318 • Fitting the same data at every rung eliminates Monte Carlo noise between
 319 models.

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327 *This is also what makes a model ladder informative. Each rung is a dif-*
328 *ferent working covariance V_0 fitted to the same data, so tracking r along a*
329 *graded sequence of working covariances, from the simplest to the production*
330 *model, isolates the rung, that is, the covariance component, at which r de-*
331 *parts from one. Because every rung is fitted to the identical simulated data,*
332 *the comparison is free of Monte Carlo noise between models.*

333 2.3 The sample-size gradient

334 **The sample-size gradient distinguishes a finite-sample artefact, which at-**
335 **tenuates with N , from an asymptotic bias, which persists, and the dis-**
336 **inction determines the appropriate remedy.**

- 337 • A finite-sample artefact diminishes as variance parameters are estimated
338 more precisely; Kenward-Roger corrections address this case.
- 339 • An asymptotic bias is a property of the working covariance mismatch and
340 does not attenuate with sample size.
- 341 • A flat calibration profile across a range of N values identifies the asymptotic
342 case.

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350 *A standard-error mis-calibration may be one of two qualitatively different*
351 *things. It may be a finite-sample artefact, in which case it attenuates as*
352 *the sample grows and the working-covariance parameters are estimated more*
353 *sharply; the small-sample corrections of Kenward and Roger [-@kenward1997;*
354 *-@kenward2009] address mis-calibration of this kind. Or it may be asymp-*
355 *totic, a consequence of a working covariance that does not match the data-*
356 *generating covariance no matter how much data is collected, in which case*
357 *it persists. The two are distinguished by running the decomposition across*
358 *a range of sample sizes. Attenuation toward nominal calibration marks the*
359 *finite-sample case; a flat profile marks the asymptotic case. The distinction*
360 *matters, because a finite-sample artefact at a realistic sample size may be*
361 *tolerable, whereas an asymptotic bias is a property of the method and will*
362 *not be outgrown.*

363 3 Worked example: an interaction test under carry- 364 over

365 **The worked example is a biomarker-by-treatment interaction test from**
366 **an aggregated N-of-1 carryover simulation study, reported in ADEMP**
367 **structure.**

- 368 • Participants are repeatedly crossed between active treatment and control
369 in an N-of-1 design.
- 370 • The scientific question is whether a baseline biomarker moderates the treat-
371 ment effect.
- 372 • Carryover complicates the analysis and the study compares strategies for
373 accommodating it.

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381 *The example we shall work through is the test of a biomarker-by-treatment*
382 *interaction in the carryover-sensitivity simulation study of the pmsimstats-*
383 *ng project. Aggregated N-of-1 trials repeatedly cross each participant between*
384 *an active treatment and a control condition, and the scientific question is*
385 *whether a baseline biomarker moderates the treatment effect. Carryover,*
386 *the persistence of drug effect into a washout period, complicates the analy-*
387 *sis, and the study compares analysis-model strategies for accommodating it*
388 *[@morris2019simulation]. The reporting follows the ADEMP structure of*
389 *Morris, White, and Crowther [-@morris2019simulation].*

390 3.1 The analysis model

391 **The analysis model is an nlme::lme with a random intercept and cor-**
392 **CAR1 residual correlation, and the calibration of its interaction test is**
393 **the question under examination.**

- 394 • The model includes a participant random intercept and a continuous-time
395 AR(1) residual correlation.
- 396 • The corCAR1 form was adopted for clinical realism and numerical tractabil-
397 ity.
- 398 • Whether it is well calibrated for the fixed-effect interaction test is the
399 central question.

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For a symptom score $S_{x,it}$ on participant i at timepoint t , the analysis model

is the linear mixed model

$$S_{x,it} = \beta_0 + \beta_1 \text{bm}_i + \beta_2 t + \beta_3 X_{it} + \theta (\text{bm}_i \times X_{it}) + b_i + \varepsilon_{it},$$

fitted by `nlme::lme` with a participant random intercept $b_i \sim N(0, \sigma_b^2)$ and a continuous-time first-order autoregressive residual correlation, `corCAR1(form = ~ t | ptID)`. Here bm_i is the standardised biomarker, X_{it} a drug-exposure predictor, and θ the interaction coefficient of interest. The continuous-time autoregressive residual was adopted in the project for its clinical realism, nearby measurements being more strongly correlated than distant ones, and for the numerical range it affords [Pinheiro2000]. Whether that choice is also well calibrated for the fixed-effect interaction test is precisely the question we shall address.

3.2 The null configuration

The null configuration collapses the two DGP architectures to a single process, so the calibration diagnosis need not stratify by architecture.

- Calibration is a null-hypothesis property, evaluated at $c_{bm} = 0$ and $\theta = 0$.
- The primary cell is the Hybrid design at $N = 70$ with a one-week carryover half-life.
- At the null, both DGP architectures produce bit-identical data.

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Calibration is a null-hypothesis property, so the diagnosis is run where the biomarker exerts no moderation, $c_{bm} = 0$, and the true interaction is therefore $\theta = 0$. The primary cell is the reference configuration of the parent study: the Hybrid N-of-1 design, $N = 70$ participants, an exponential carryover form with a one-week half-life, and an autoregressive parameter of 0.7. A preliminary check established that at $c_{bm} = 0$ the study's two data-generating architectures collapse to a bit-identical process, so the diagnosis need not stratify by architecture.

The working covariance is structurally incapable of matching the data-generating covariance, making this a prototypical instance of the misspecification problem.

- The simulated symptom score is a mixture of three autoregressive factors with timepoint-dependent variances plus a participant baseline.
- A single random intercept and single-parameter `corCAR1` residual cannot reproduce this structure.
- The example is therefore one of plausible, conventional, and inexact working covariance specification.

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*The feature of the example that merits attention for the argument is the
relationship between the working covariance and the truth. The simulated
symptom score is a sum of three latent factors, each following its own autore-
gressive process, with timepoint-dependent variances, plus a participant-level
baseline term. The working covariance of the analysis model, one random
intercept and one single-parameter corCAR1 residual, cannot reproduce that
structure exactly. The example is therefore an instance of the general situ-
ation the protocol is designed for: a parametric working covariance that is
plausible, conventional, and not exactly correct.*

4 Results

**Simulation runs are sized to keep Monte Carlo standard error below 0.005
for each diagnostic component, with all replicate-level outputs retained.**

- The decomposition and model ladder use 5000 null replicates; the sample-
size gradient uses 1500 per point.
- The cluster-robust verification uses 3000 replicates.
- At a type-I error near 0.03, the Monte Carlo standard error is approximately
0.0024 at 5000 replicates.

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*The diagnosis was carried out at 5000 null replicates for the four-channel
decomposition and the model ladder, and at 1500 replicates per point for
the sample-size gradient. The cluster-robust remedy was verified at 3000
replicates. A rejection rate estimated from n_{sim} replicates carries a Monte
Carlo standard error of $\sqrt{p(1-p)/n_{\text{sim}}}$ [atmorris2019simulation]; at a type-
I error near 0.03 this is approximately 0.0024 at 5000 replicates, 0.0031 at
3000, and 0.0044 at 1500, and the comparisons to the nominal level reported
below are read against those tolerances. All replicate-level outputs are retained
for inspection.*

4.1 The conservatism is standard-error over-estimation

**The decomposition table excludes point-estimate bias and the degrees-
of-freedom channel, leaving the standard-error channel as the site of the
conservatism.**

Table 1: Four-channel decomposition at the primary null cell (Hybrid design, $N = 70$, $c_{bm} = 0$, 5000 replicates), for the three analysis specifications A1, A2, A3. The interaction estimate is effectively unbiased; the SE ratio exceeds one; the statistic is correspondingly compressed; the t and z references coincide because the degrees of freedom are large.

Spec	Mean estimate	SE ratio r	SD(T)	Type I (t)	Type I (z)	Mean DF
A1	+0.032	1.069	0.932	0.033	0.034	487
A2	+0.039	1.097	0.909	0.029	0.029	487
A3	+0.033	1.061	0.939	0.035	0.036	486

- All three specifications show a negligible null-mean displacement of about 0.035 standard deviations.
- The model assigns approximately 487 degrees of freedom; switching to a normal reference moves the type-I error by only 0.001.
- The conservatism must therefore reside in the standard-error channel.

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Table
reftab:tab-decomp reports the decomposition for the three analysis specifications carried by the parent study, which differ only in the drug-exposure predictor X_{it} and are treated here as three instances of the same interaction test. The reading is consistent across all three. The interaction estimate is effectively unbiased: the null mean falls between +0.03 and +0.04 across the three specifications, on a coefficient whose sampling standard deviation is near unity, a displacement of about 0.035 of a standard deviation. A displacement of that size is statistically detectable at 5000 replicates but is negligible in magnitude, and in any case it points toward rejection and so cannot produce a conservative test. The degrees-of-freedom channel is excluded directly: the model assigns approximately 487 degrees of freedom to the interaction term, and recomputing the p -values against a normal reference moves the realised type-I error by about 0.001, as the near-identical t - and z -reference columns show.

The discrepancy is in the standard-error channel. The model-based standard error exceeds the empirical sampling standard deviation of the estimate by 6 to 10 percent, and the standard deviation of the test statistic agrees with the reciprocal of that ratio to within half a percent, as the identity $SD(T) \approx SD(Z^*)/r$ with $SD(Z^*) = 1$ leads us to expect. With the other three channels inert, $\delta \approx 0$, ν large, and Z^* standard normal, the realised type-I error of Section 2.1 reduces to

524 the closed form

$$\alpha_{\text{real}} \approx 2 \Phi(-r t_{\nu, 1-\alpha/2}),$$

525 the probability that a standard normal exceeds the threshold $r t_{\nu, 1-\alpha/2}$ rather
526 than $z_{1-\alpha/2}$.

527 **A first-order approximation shows the realised type-I error falls 2.3 times**
528 **as fast as the standard error is inflated, so a 6–10 percent over-statement**
529 **is materially consequential.**

- 530 • With the other three channels inert the realised error reduces to
531 $2\Phi(-r t_{\nu, 1-\alpha/2})$.
- 532 • The derivative near $r = 1$ is approximately -0.23 , so the type-I error falls
533 to the low 0.03 range.
- 534 • The predicted range of 0.031 to 0.037 matches the observed 0.029 to 0.035
535 to within Monte Carlo error.

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Differentiating this closed form exposes the leverage of the scale channel.
Near $r = 1$, with ν large enough that $t_{\nu, 1-\alpha/2} \approx z_{1-\alpha/2} = 1.96$,

$$\frac{d \alpha_{\text{real}}}{d r} = -2 t_{\nu, 1-\alpha/2} \phi(r t_{\nu, 1-\alpha/2}) \approx -0.23,$$

543 *so to first order $\alpha_{\text{real}} \approx 0.05 - 0.23(r - 1)$: the realised error falls roughly*
544 *2.3 times as fast as the standard error is inflated. A six-to-ten percent*
545 *over-statement of the standard error therefore depresses the type-I error into*
546 *the low 0.03 range, and it is in this sense that a numerically modest mis-*
547 *calibration of the standard error constitutes a material mis-calibration of the*
548 *test. For the standard-error ratios of Table*
549 *reflab:tab-decomp this predicts a type-I error of 0.031 to 0.037 across the three*
550 *specifications, matching the observed 0.029 to 0.035 (at $\alpha = 0.05$) to within*
551 *Monte Carlo error. The scale channel alone, with no contribution from the*
552 *other three, reproduces the conservatism.*

553 **The uniform bowing of the p-value CDF below the diagonal confirms**
554 **global scale compression rather than a tail anomaly, and the positive-**
555 **definiteness correction is excluded as a contributor.**

- 556 • A tail-only failure would depart from the diagonal only near zero; the uni-
557 form bowing seen here is distinctive.
- 558 • Global scale compression is the signature of an inflated standard error.
- 559 • The positive-definiteness correction was not triggered in any of the cell's
560 replicates.

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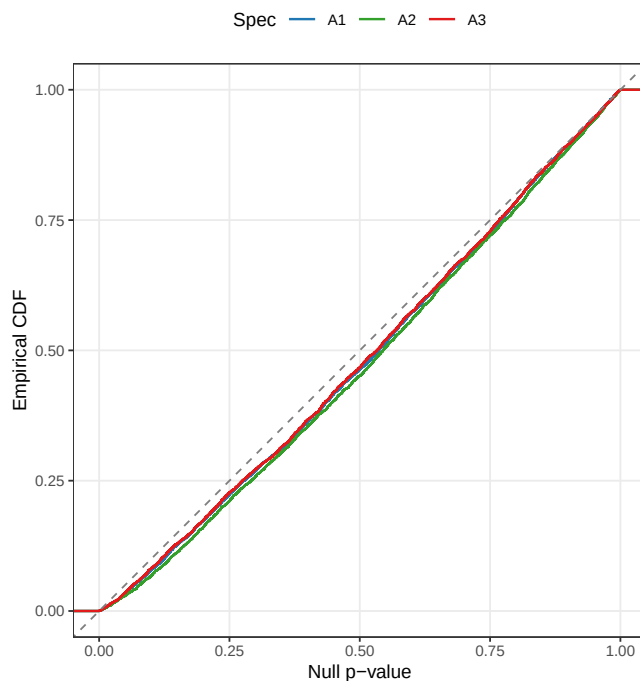


Figure 1: Empirical distribution function of the null p-values at the primary cell, against the Uniform(0,1) diagonal. All three specifications bow uniformly below the diagonal, the signature of a global scale compression rather than a tail anomaly.

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Figure

reffig:fig-ecdf confirms the nature of the mis-calibration. The empirical distribution function of the null p-values lies below the diagonal across the whole unit interval and for all three specifications. A test that failed only in its tails would depart from the diagonal only near zero; the uniform bowing seen here is the signature of a global scale compression, which is precisely what an inflated standard error produces. The positive-definiteness correction available to the data generator was not triggered in any of the cell's paths and is excluded as a contributor.

4.2 The corCAR1 residual layer is the cause

The model ladder identifies the source of the over-estimation by fitting four working covariances to the identical null datasets.

- M0 is OLS with no within-cluster term; M1 adds a random intercept; M2 adds only corCAR1; M3 combines both.
- Because every rung uses the same data, the comparison is free of Monte Carlo noise.
- The rung at which r departs from one identifies the responsible component.

Table 2: Working-covariance model ladder at the primary null cell (5000 replicates), for specifications A1 and A2. The same datasets are refitted under each structure. M1, a random intercept alone, is well calibrated; M3, the production model, adds a corCAR1 residual and is conservative.

Model	Structure	r (A1)	r (A2)	Type I (A1)	Type I (A2)
M0	OLS, no within-cluster correlation	1.323	1.289	0.010	0.012
M1	Random intercept only	0.989	0.972	0.049	0.055
M2	corCAR1 residual only	1.023	1.070	0.043	0.036
M3	Random intercept + corCAR1 (production)	1.068	1.097	0.033	0.029

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592 *The decomposition tells us that the standard error is over-estimated. The*
593 *model ladder, Table*
594 *refstab:tab-ladder, tells us which part of the model does the over-estimating.*
595 *Four working covariances were fitted to the identical null datasets: ordinary*
596 *least squares with no within-cluster term (M0), a random intercept alone*
597 *(M1), a corCAR1 residual alone (M2), and the production combination of a*
598 *random intercept with a corCAR1 residual (M3).*

599 **The corCAR1 residual layer is the proximate cause: adding it to a random**
600 **intercept that is itself well calibrated removes the calibration.**

- 601 • M1 yields SE ratios of 0.97–0.99 and type-I errors of 0.049–0.055.
- 602 • M3 raises the ratio to 1.07–1.10 and depresses the type-I error to 0.029–
- 603 0.033.
- 604 • The claim is not that corCAR1 is wrong absolutely but that adding it to a
- 605 model that already carries a random intercept degrades calibration.

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613 *The result is unambiguous. M1, the random intercept with no residual corre-*
614 *lation, is well calibrated: its standard-error ratio sits between 0.97 and 0.99*
615 *and its type-I error between 0.049 and 0.055. M3, which adds the corCAR1*
616 *residual on top of that random intercept, is conservative: the ratio rises to*
617 *between 1.07 and 1.10 and the type-I error falls to between 0.029 and 0.033.*
618 *The corCAR1 residual-correlation layer is therefore the proximate cause of*

the conservatism. It is worth being precise about what this does and does not say. It does not say that the *corCAR1* residual is wrong in any absolute sense; the simulated factors are genuinely autocorrelated, so a residual correlation term is capturing a real feature of the data. What it says is that adding that term to a model which already carries a random intercept degrades the calibration of the fixed-effect interaction test.

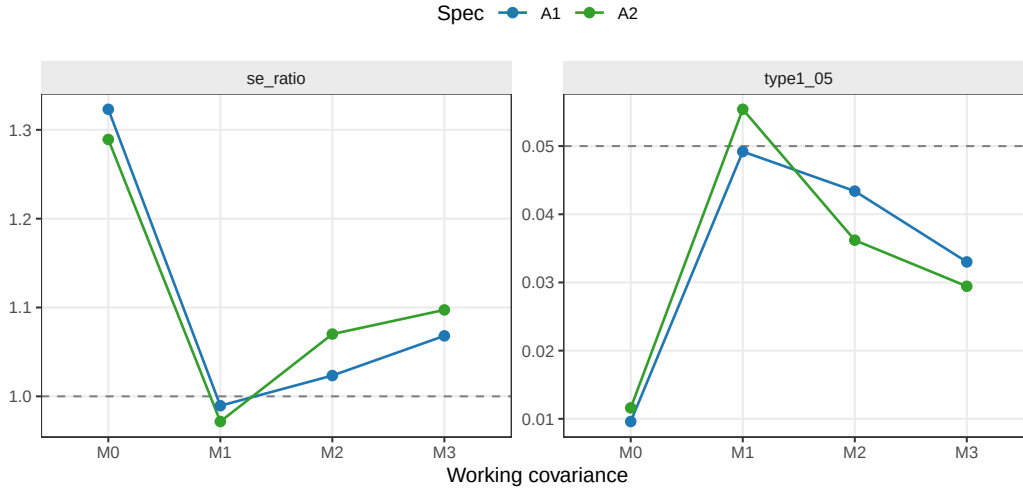


Figure 2: Interaction standard-error calibration across the model ladder. Left, the ratio of model-reported to empirical standard error; right, the realised type-I error. The dashed lines mark perfect calibration. The random intercept alone (M1) sits on both reference lines; the production model (M3) does not.

OLS is strongly conservative for within-cluster contrasts – reversing the usual Moulton intuition – and the ladder shows that the sign of the OLS error is a property of the estimand, not of ignoring correlation per se.

- OLS charges the full residual variance to a within-participant contrast when only the within-participant component is relevant, yielding $r \approx 1.3$.
- The Moulton pitfall applies to between-cluster effects; for within-cluster contrasts the sign is reversed.
- M2 (*corCAR1* alone) is intermediate, confirming that mis-calibration magnitude and sign depend on the working covariance choice.

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Two further features of the ladder, visible in Figure *reffig:fig-ladder*, are worth recording. The first is that M0, ordinary least squares, is not anti-conservative, as the familiar warning that ignoring positive within-cluster correlation under-states standard errors would suggest. It is strongly conservative, with a standard-error ratio near 1.3. The reason is instructive: the interaction here is a within-participant contrast, and ordinary

Table 3: Sample-size gradient under the production model M3 (1500 replicates per point). The standard-error ratio r and the type-I error (T1) are shown for each specification. Neither trends toward its nominal value as N increases.

N	r A1	r A2	r A3	T1 A1	T1 A2	T1 A3
35	1.072	1.103	1.064	0.038	0.028	0.039
70	1.056	1.083	1.050	0.043	0.031	0.045
140	1.086	1.098	1.079	0.041	0.033	0.042
280	1.052	1.070	1.046	0.037	0.028	0.035

least squares charges it the full residual variance, between-participant and within-participant combined, when the error relevant to a within-participant contrast is only the within-participant part. In the idealised case of a pure within-participant contrast with exchangeable errors the inflation is the variance ratio $r_{OLS}^2 \approx (\sigma_b^2 + \sigma_\varepsilon^2)/\sigma_\varepsilon^2$, the total residual variance over its within-participant component, exceeding one by the random-intercept variance σ_b^2 . The familiar warning, the Moulton pitfall, concerns between-cluster effects [Moulton1990]; for a within-cluster effect the sign of the OLS error is reversed. The second feature is that the corCAR1 residual without a random intercept (M2) is intermediate. Taken together, the ladder shows that calibration is recovered by the random intercept alone and removed by the corCAR1 addition, and that the magnitude and even the sign of the mis-calibration is a property of the working covariance, not a universal constant.

4.3 The conservatism is asymptotic, not a small-sample artefact

The conservatism is asymptotic: the SE ratio and type-I error show no trend toward their nominal values across an eightfold range of sample size.

- From $N = 35$ to $N = 280$ the SE ratio holds between 1.05 and 1.10 with no downward trend.
- At $N = 280$ the type-I error is still 0.028, well below the nominal 0.05.
- The conservatism is a property of the M3 working covariance and will not be outgrown.

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If the conservatism were a finite-sample artefact, it would diminish as the sample grew and the working-covariance parameters were estimated more precisely. Table reftab:tab-grad and Figure reffig:fig-grad show that it does not. Across an eightfold range of sample size, from $N = 35$ to $N = 280$ participants, the standard-error ratio holds between

1.05 and 1.10 with no trend toward one, and the type-I error holds between 0.028 and 0.045 with no trend toward 0.05. At the largest sample size the production specification still rejects the true null at 0.028. The conservatism is asymptotic. It appears to be a property of the M3 working covariance and will not be outgrown.

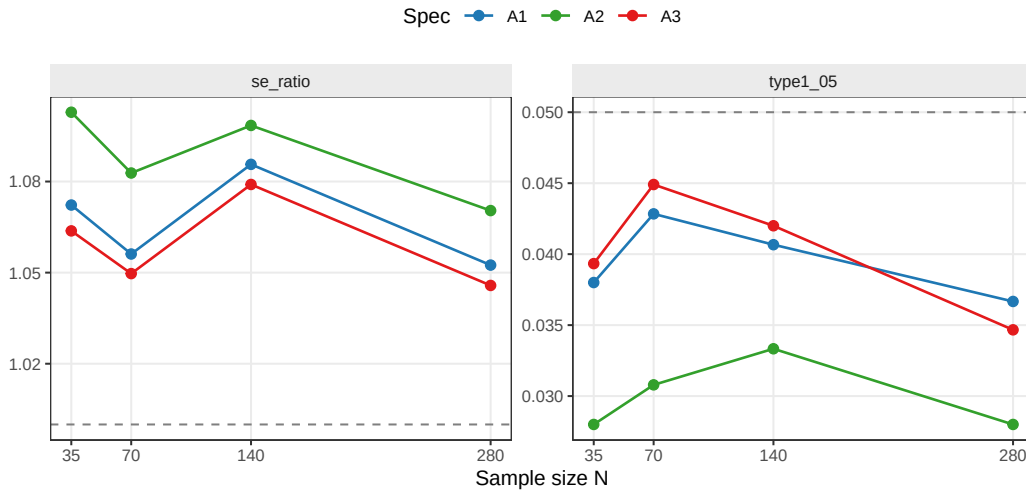


Figure 3: Type-I error and standard-error ratio against sample size under the production model M3. Neither panel trends toward its dashed reference line as the number of participants increases from 35 to 280.

The asymptotic bias arises because the model-based SE converges to a biased target: more data sharpens estimation of the wrong covariance, not the right one.

- The data-generating covariance is a mixture of autoregressive factors with timepoint-dependent variances that M3 cannot reproduce.
- Larger samples estimate the M3 parameters more precisely but do not correct the structural mismatch.
- The bias is a property of the model, not of the estimation of the model's parameters.

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The explanation that is consistent with all three results is a working-covariance mismatch rather than a small-sample effect. The simulated symptom score is a mixture of autoregressive factors with unequal, timepoint-dependent variances, plus a participant baseline term. The M3 working covariance, a single random intercept and a single-parameter *corCAR1* residual, cannot match a mixture of autoregressive processes exactly. Increasing the number of participants estimates the parameters of that working covariance ever more sharply, but it does not change the fact

Table 4: Cluster-robust remedy at the primary null cell (3000 replicates). The model-based standard error of the production model is replaced by a CR2 bias-reduced cluster-robust standard error clustered on participant, with Satterthwaite degrees of freedom. The CR2 standard-error ratio is approximately one and the type-I error approximately 0.05.

Spec	r model-based	r CR2	Type I model	Type I CR2	CR2 df
A1	1.062	0.978	0.036	0.048	23.7
A2	1.096	1.000	0.029	0.045	23.7
A3	1.053	0.976	0.039	0.049	23.7

that the covariance being estimated is, in a sense, the wrong one, and so the model-based standard error converges to a biased target. The bias persists because it is a bias of the model, not of the estimate of the model's parameters.

4.4 A cluster-robust standard error restores calibration

The natural remedy is a cluster-robust SE that is consistent regardless of the working covariance; the CR2 bias-reduced form is chosen for the moderate-cluster regime.

- The corCAR1 model and its point estimate are retained; only the standard error is replaced.
- The CR2 estimator of Bell and McCaffrey targets the moderate-cluster setting and is implemented in `clubSandwich`.
- Satterthwaite degrees of freedom from Pustejovsky and Tipton provide the matching small-sample reference.

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The diagnosis points to a remedy that does not require abandoning the corCAR1 model. If the model-based standard error is biased because the working covariance is wrong, the natural replacement is a standard error that is consistent whatever the working covariance happens to be [white1980; liang1986; royall1986]. We replace the model-based standard error of the interaction with a cluster-robust sandwich standard error, clustered on participant, in the bias-reduced CR2 form of Bell and McCaffrey [bell2002], with the Satterthwaite degrees of freedom of Pustejovsky and Tipton [pustejovsky2018; satterthwaite1946]. The CR2 adjustment and its associated small-sample test were designed for precisely the regime we face here, a moderate number of clusters, and are implemented for `lme` objects in the `clubSandwich` package [tipton2015; tipton2015moderators].

744 The CR2 estimator uses adjustment matrices $A_i = (I - H_{ii})^{-1/2}$ to achieve
 745 exact unbiasedness under the working covariance, and Satterthwaite de-
 746 grees of freedom are recovered from the eigenvalues of the defining
 747 quadratic form.

- 748 • CR0 is consistent but downward-biased at moderate cluster counts; the
 749 jackknife CR3 over-corrects.
- 750 • CR2 takes the symmetric square root of $(I - H_{ii})$, generalising the HC2
 751 estimator of MacKinnon and White.
- 752 • For lme objects the construction is applied in the GLS-transformed metric
 753 following Pustejovsky and Tipton.

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It is worth setting down the construction. Let $\hat{e}_i$ collect the residuals of cluster i from the fitted model and let H_{ii} be the corresponding block of the hat matrix. A cluster-robust variance replaces the model-based variance of $\hat{\theta}$ by a quadratic form in the empirical cluster cross-products,

$$\hat{V}_{\text{CR}} = (X'X)^{-1} \left(\sum_i X_i' A_i \hat{e}_i \hat{e}_i' A_i X_i \right) (X'X)^{-1},$$

761 and the family CR0, CR1, CR2, CR3 is indexed by the adjustment matrices A_i
 762 [[@cameron2015](#); [@pustejovsky2018](#)]. The unadjusted choice $A_i = I$ is consis-
 763 tent as the number of clusters grows but biased downward at moderate cluster
 764 counts; the jackknife choice $A_i = (I - H_{ii})^{-1}$ over-corrects and is, in the
 765 marginal setting, the estimator of Mancl and DeRouen [[@mancl2001](#)]. The
 766 CR2 choice sets $A_i = (I - H_{ii})^{-1/2}$, the symmetric square root that renders
 767 $\hat{V}_{\text{CR}}$ exactly unbiased for the true variance when the working covariance is in
 768 fact correct; it is the cluster generalisation of the HC2 estimator of MacK-
 769 innon and White [[@mackinnon1985](#)]. For a mixed model fitted by lme the
 770 construction is applied in the metric of the working covariance, that is, to the
 771 generalized-least-squares-transformed design and residuals, following Puste-
 772 jovsky and Tipton [[@pustejovsky2018](#)], and is recommended with a matching
 773 small-sample reference by Imbens and Kolesar [[@imbens2016](#)]. Because $\hat{V}_{\text{CR}}$
 774 is itself a quadratic form in approximately normal residuals, its null distribu-
 775 tion is approximated by a scaled χ^2 , and the Satterthwaite degrees of freedom
 776 are recovered by matching its first two moments, $\nu = (\sum_g \lambda_g)^2 / \sum_g \lambda_g^2$ in the
 777 eigenvalues λ_g of the matrix defining that form [[@satterthwaite1946](#); [@puste-](#)
 778 [jovsky2018](#)].

779 The CR2 test restores the SE ratio to approximately one and the type-I
 780 error to approximately 0.05; Satterthwaite degrees of freedom of approx-
 781 imately 24 confirm the small-sample reference is doing real work.

- 782 • CR2 SE ratios are 0.97–1.00 versus 1.05–1.10 for the model-based test.
- 783 • The type-I error rises from approximately 0.03 to 0.045–0.049, within

Monte Carlo error of 0.05.

- Satterthwaite df of approximately 24 (versus model-based 487) indicates that a naive normal reference would be mildly anti-conservative.

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Table

reftab:tab-cr2 reports the verification. The model-based standard-error ratio reproduces the conservatism, between 1.05 and 1.10. The CR2 cluster-robust ratio is close to one, between 0.97 and 1.00, and the realised type-I error rises from approximately 0.03 under the model-based test to between 0.045 and 0.049 under the CR2 test, within Monte Carlo error of the nominal 0.05. The Satterthwaite degrees of freedom for the cluster-robust test are approximately 24, far below the model-based 487 and well below the participant count of 70, since they measure the heterogeneity of cluster leverage on the interaction contrast rather than a count of clusters [imbens2016]; the small-sample reference is doing real work, and a naive normal reference for the sandwich statistic would itself be mildly anti-conservative [cameron2015; long2000].

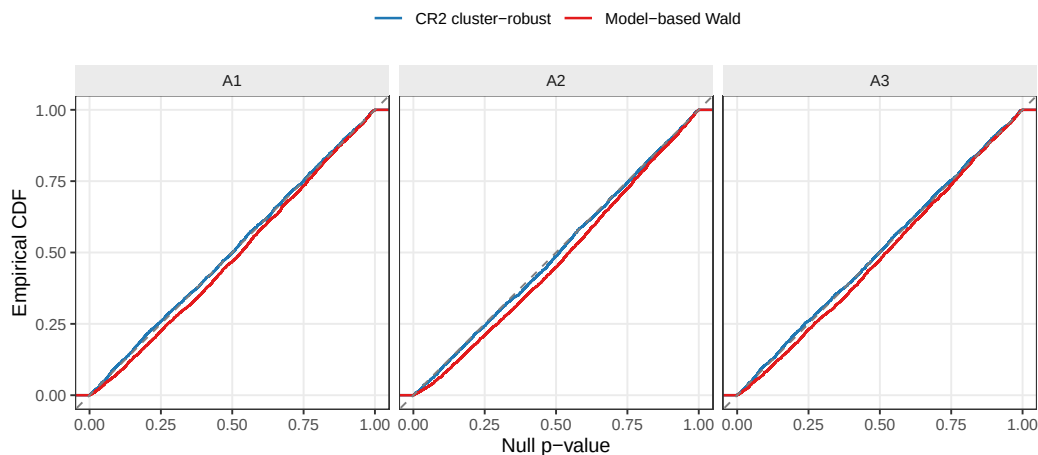


Figure 4: Null p-value distribution functions under the model-based and CR2 cluster-robust tests, by specification. The model-based curves bow below the diagonal; the CR2 curves lie on it.

The p-value CDF confirms that CR2 restores calibration without altering the point estimate or the analysis model.

- Model-based p-values bow below the diagonal consistently across all three specifications.
- Cluster-robust p-values lie on the Uniform(0,1) diagonal.
- Only the standard error and its reference distribution are replaced; the fitted model is otherwise unchanged.

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820 *Figure*
821 *reffig:fig-cr2 shows the same result as a distribution function. The model-*
822 *based p-values bow below the diagonal, as in Figure*
823 *reffig:fig-ecdf; the cluster-robust p-values lie on it. The point estimate and*
824 *the corCAR1 model are unchanged; only the standard error of the interaction,*
825 *and the reference against which it is tested, are replaced.*

826 4.5 The remedy at an alternative cell

827 **The comparative ranking of specifications is checked at one alternative**
828 **cell to determine whether recalibration disturbs their order.**

- 829 • The alternative cell uses $c_{bm} = 0.45$, the parent study's reference non-null
830 configuration.
- 831 • The question is whether recalibration changes not only absolute power but
832 the ordering of specifications.
- 833 • The Architecture-B Hybrid design at $N = 70$ is retained for comparability
834 with the null diagnosis.

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842 *The diagnosis to this point has been conducted under the null, where calibra-*
843 *tion is defined. A reader of a comparative study will, however, press a further*
844 *question. The purpose of the parent study is to rank analysis specifications*
845 *by power, and if a recalibration of the standard error raises the power of*
846 *every specification, does it also disturb their order? We checked this directly*
847 *at one alternative cell, the parent study's reference configuration with a non-*
848 *null biomarker moderation, $c_{bm} = 0.45$, retaining the Architecture-B Hybrid*
849 *design at $N = 70$.*

850 **The recalibration acts as a near-uniform level shift that raises absolute**
851 **power without disturbing the ranking of specifications.**

- 852 • At the alternative cell the model-based ratio is 1.12–1.15; CR2 brings it to
853 0.99 for all three specifications.
- 854 • Power rises by roughly three to four percentage points across specifications.
- 855 • The gaps between specifications are preserved: A2's advantage over A1
856 and A3 is essentially unchanged.

Table 5: The cluster-robust remedy at an alternative cell (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, 3000 replicates). Power is the rejection rate of the interaction test; the standard-error ratio r is as defined for the null. The model-based test is conservative here too, and the CR2 test recalibrates it.

Spec	r model-based	r CR2	Power model	Power CR2
A1	1.124	0.986	0.757	0.795
A2	1.152	0.988	0.843	0.878
A3	1.119	0.989	0.755	0.788

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Three things follow from Table reftab:tab-alt. First, the standard-error over-estimation is not confined to the null: at the alternative cell the model-based ratio is 1.12 to 1.15, if anything slightly larger than the 1.06 to 1.10 seen under the null. Second, the CR2 cluster-robust standard error recalibrates the alternative cell as it did the null, bringing the ratio to 0.99 for all three specifications. Third, and most consequential for a comparative study, the recalibration is close to a uniform level shift. It raises the power of every specification by roughly three to four percentage points, and the gaps between specifications are left intact: the advantage of the leading specification A2 over A1 is +0.086 under the model-based test and +0.082 under the cluster-robust test, and over A3 it is +0.088 and +0.089. The ranking, and the magnitude of the leading specification's advantage, survive the recalibration unchanged.

A comparative study whose claim is a ranking is robust to recalibration because the correction shifts the level without reordering the contrast.

- The cluster-robust correction acts on the standard error, not on the point estimate.
- Because the mis-calibration is similar across specifications, the shift is approximately uniform.
- Absolute power figures reported under the model-based test are mildly pessimistic but the ranking is unaffected.

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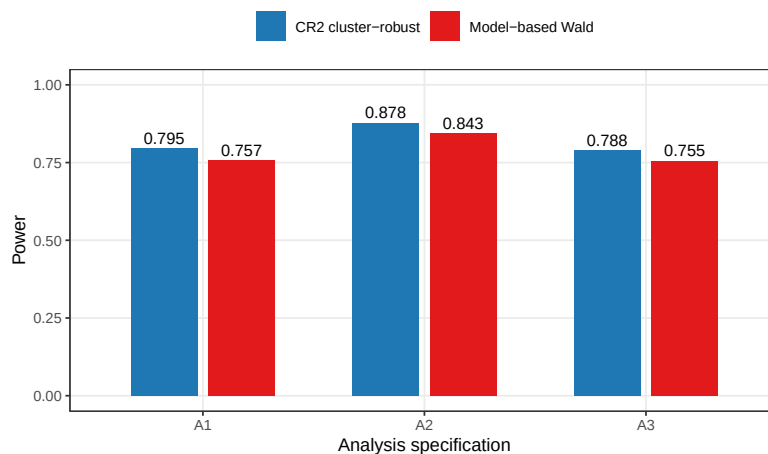


Figure 5: Power of the three analysis specifications at the alternative cell, under the model-based and CR2 cluster-robust tests. The recalibration raises every specification by a near-uniform increment and preserves their order.

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892 *The lesson generalises beyond the example. A cluster-robust standard error*
 893 *corrects the absolute error rate, the type-I error under the null and the power*
 894 *under the alternative alike, but because the correction acts on the standard*
 895 *error and not on the point estimate, and because the standard-error mis-*
 896 *calibration here is relatively similar across the specifications being compared,*
 897 *it shifts the level without reordering the contrast. A comparative study whose*
 898 *claim is a ranking is therefore robust to the recalibration, even where its*
 899 *absolute power figures, as reported under the model-based test, are mildly*
 900 *pessimistic.*

901 5 Discussion

902 5.1 The result is not about autoregressive correlation

903 **The mechanism is general: any parametric working covariance capable of**
 904 **mismatching the truth is capable of mis-calibrating the interaction test.**

- 905 • The corCAR1 term is the component implicated here, but the model ladder
- 906 shows four different ratios for four different working covariances.
- 907 • Mis-calibration from a wrong working covariance is the Eicker-Huber-White
- 908 result; the worked example adds quantification and decomposition.
- 909 • The specific contribution is a realistic longitudinal case with two non-
- 910 obvious features.

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918 *It would be a mistake to read this paper as a caution against the continuous-*
919 *time autoregressive residual in particular. The corCAR1 term is the compo-*
920 *nent the model ladder happened to implicate in this example, but the mecha-*
921 *nism is general. The model ladder is itself the evidence: it fitted four working*
922 *covariances and obtained four different standard-error ratios, from 1.3 under*
923 *ordinary least squares to 1.0 under a random intercept to near 1.1 under the*
924 *production model. The mis-calibration is a property of the match between the*
925 *working covariance and the truth, and any parametric residual or random-*
926 *effects structure, compound symmetry, an unstructured form, a random slope,*
927 *a spatial correlation, is capable of producing it. That a mis-specified work-*
928 *ing covariance leaves the model-based fixed-effect standard error inconsistent*
929 *is, moreover, not a new claim; it is the Eicker-Huber-White result carried*
930 *into the mixed-model setting [Eicker1967; Huber1967; White1980; Freed-*
931 *man2006]. What the worked example contributes is a quantified, decomposed*
932 *instance in a realistic longitudinal interaction test, with two features that are*
933 *not obvious in advance.*

934 5.2 Two features that are not obvious in advance

935 **For a within-cluster contrast the direction of the standard-error bias**
936 **can be conservative rather than anti-conservative, reversing the usual**
937 **sandwich-literature intuition.**

- 938 • The sandwich literature is typically motivated by anti-conservative standard
939 errors from ignored between-cluster correlation.
- 940 • Here the model-based test is conservative: the corCAR1 term, not the
941 absence of modelling, introduced the bias.
- 942 • An analyst who infers conservatism from having modelled the correlation
943 has the logic backward for a within-cluster contrast.

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951 *The first is the direction. The sandwich literature is usually motivated by*
952 *anti-conservative tests: ignored positive correlation under-states standard er-*
953 *rors, and the robust estimator corrects an inflated rejection rate downward.*
954 *Here the model-based test is conservative, and the cluster-robust estimator*
955 *corrects the rejection rate upward. The ladder explains why the intuition can*
956 *mislead. The effect under test is a within-cluster contrast, and for a within-*
957 *cluster contrast the relationship between a mis-specified covariance and the*
958 *standard-error bias need not have the sign that the between-cluster intuition*
959 *supplies. An analyst who reasons that 'my standard errors are conservative*
960 *because I have modelled the correlation' has the logic backward in this case:*
961 *modelling the correlation, through the corCAR1 term, is what introduced the*
962 *conservatism, and the less elaborate model was the calibrated one.*

963 **Kenward-Roger corrections address finite-sample artefacts and are the**

964 **wrong instrument here because the bias is asymptotic and does not at-**
965 **tenuate with sample size.**

- 966 • An earlier informal reading attributed the conservatism to moderate sample
967 size; the sample-size gradient refutes this.
- 968 • Kenward-Roger corrections operationalise the Kackar-Harville result for
969 finite-sample variance underestimation.
- 970 • An asymptotic mismatch is not corrected by Kenward-Roger; the cluster-
971 robust SE addresses it directly.

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979 *The second feature is that the bias is asymptotic. It would have been straight-*
980 *forward to attribute the conservatism to moderate sample size, and an ear-*
981 *lier informal reading of the parent study did so. The sample-size gradient*
982 *refutes that reading directly. This matters for how the problem is classi-*
983 *fied. The small-sample corrections of Kenward and Roger [-@kenward1997;*
984 *-@kenward2009] adjust the fixed-effect standard error and its degrees of free-*
985 *dom for the finite-sample uncertainty in the estimated variance parameters,*
986 *operationalising the variance-underestimation result of Kackar and Harville*
987 *[-@kackar1984]; they are the right instrument for a finite-sample artefact.*
988 *They are not the right instrument here, because the bias is not a finite-sample*
989 *artefact: it does not vanish as the variance parameters are estimated with*
990 *unlimited precision. A Kenward-Roger correction applied to a mis-specified*
991 *working covariance corrects the finite-sample term and leaves the asymptotic*
992 *mismatch untouched. The cluster-robust standard error addresses the asymp-*
993 *totic mismatch directly, which is why it is the appropriate remedy.*

994 5.3 When the problem should be expected

995 **The conditions for this problem are met by nearly every applied mixed-**
996 **model interaction test, and routine accompaniment by a calibration check**
997 **or cluster-robust SE is warranted.**

- 998 • The conditions are: a fixed-effect interaction, a parametric working covari-
999 ance not known to be exact, and a within-cluster contrast.
- 1000 • These conditions characterise almost every applied mixed-model interaction
1001 analysis.
- 1002 • A calibration check or cluster-robust SE removes the question at modest
1003 additional cost.

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The conditions for the problem are not exotic. They are a fixed-effect interaction tested in a mixed model; a parametric working covariance that is plausible but not known to be exactly correct, which is to say almost every parametric working covariance in applied use; and an interaction that is partly or wholly a within-cluster contrast. Wherever those conditions hold, the calibration of the model-based Wald test is an open question, and the staged protocol of Section 2 answers it at the cost of a simulation. We would suggest, as a matter of routine practice, that an interaction test reported from a mixed model with a non-trivial working covariance be accompanied either by a calibration check of this kind or by a cluster-robust standard error, which removes the question by being agnostic to the working covariance. This accords with a broader case for giving robust and marginal methods more routine consideration alongside the conditional mixed model [mcneish2017].

5.4 The marginal-model alternative

GEE is a natural alternative that carries the sandwich variance from the outset; for a linear model with identity link it estimates the same interaction as the conditional mixed model.

- Liang-Zeger GEE decouples point estimation from variance estimation by design.
- For a linear identity-link model, marginal and conditional fixed-effect coefficients coincide.
- The working correlation governs efficiency, not inferential validity.

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The remedy adopted above keeps the conditional mixed model and adds a cluster-robust standard error after the fact. A reader may reasonably ask whether the marginal route, generalized estimating equations, would not be more natural, since the method carries the sandwich variance from the outset rather than bolting it on [liang1986]. It would. For a linear model with an identity link the marginal and conditional fixed-effect coefficients coincide, so a generalized estimating equations analysis estimates the same interaction and answers the same question; the working correlation, exchangeable or autoregressive, governs only the efficiency of the point estimate and not the validity of the inference.

The GEE route is viable but requires its own small-sample correction; both the conditional CR2 and marginal Mancl-DeRouen tests are defensible calibrated analyses and the choice is practical rather than principled.

- The naive GEE sandwich under-estimates the sampling SD by about four

percent, yielding a type-I error of 0.067–0.069.

- Mancl-DeRouen bias correction brings the ratio to within one percent of one and the type-I error to 0.055–0.058.
- The conditional CR2 test is slightly conservative; the marginal Mancl-DeRouen test slightly anti-conservative; both are defensible.

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We checked the marginal route at the primary null cell, fitting the generalized estimating equations counterpart of the analysis model with an AR(1) working correlation. The naive sandwich standard error proved mildly anti-conservative, the small-sample behaviour the literature describes for a moderate number of clusters: across the three specifications it under-estimated the sampling standard deviation by roughly four percent, and the realised type-I error was 0.067 to 0.069. A bias-corrected sandwich of the kind introduced by Mancl and DeRouen [-@mancl2001] removed most of the discrepancy, bringing the standard-error ratio to within one percent of one and the type-I error to 0.055 to 0.058. The marginal route is therefore viable, but it carries the same lesson as the conditional one: at the cluster counts typical of aggregated N-of-1 work the sandwich variance itself needs a small-sample correction. The two corrected tests err in opposite and relatively modest directions, the conditional CR2 test slightly conservative and the marginal Mancl-DeRouen test slightly anti-conservative, and either is a defensible calibrated analysis. The choice between them is practical rather than principled: the conditional route preserves an existing mixed-model pipeline and changes only the standard error, while the marginal route is conceptually cleaner but a more thoroughgoing change of model.

5.5 Limitations

The scope is bounded to one data-generating family, one estimand, and one working covariance, and the power consequence is verified at only a single alternative cell.

- Generality beyond the corCAR1-plus-random-intercept model rests on the model ladder and established theory, not on an empirical sweep.
- The CR2 remedy spends precision (Satterthwaite df approximately 24) to buy validity, and that cost rises at smaller cluster counts.
- Whether the near-uniform power shift holds across the full factorial rather than the one cell examined remains an open question.

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1102 *Several limitations bound the scope of the evidence, and we shall endeavour*
1103 *to state them plainly. The diagnosis was carried out for one data-generating*
1104 *family, one estimand, and one production working covariance; the asymptotic*
1105 *conservatism is demonstrated for the corCAR1-plus-random-intercept model,*
1106 *and the generality beyond that rests on the model ladder and on established*
1107 *theory rather than on a sweep of every covariance structure. A broader em-*
1108 *pirical base, fitting an unstructured or compound-symmetry analysis to the*
1109 *same data, would strengthen the general claim, and is a natural extension.*
1110 *The cluster-robust remedy is not without cost: the Satterthwaite degrees of*
1111 *freedom for the CR2 test are approximately 24 against a model-based 487,*
1112 *so the robust test spends precision to buy validity, and at a genuinely small*
1113 *number of clusters that price would rise. The principal diagnosis was con-*
1114 *ducted at the null, where calibration is defined; the consequence for power*
1115 *was checked at a single alternative cell (Section 4.5), where the recalibration*
1116 *proved a near-uniform level shift that preserved the ranking of specifications.*
1117 *Whether that uniformity holds across the whole factorial, rather than at the*
1118 *one cell examined, was not established, and that question warrants further*
1119 *investigation. Finally, the small positive point-estimate bias noted in Section*
1120 *4.1, while negligible in magnitude and irrelevant to the conservatism, is a*
1121 *genuine finite-sample feature of the estimator that a separate examination*
1122 *might characterise.*

1123 6 Conclusions

1124 In conclusion, six points are to be emphasised. First, the model-based Wald test of
1125 a fixed-effect interaction in a linear mixed model can be materially mis-calibrated
1126 when the parametric working covariance does not match the data-generating co-
1127 variance. In the worked example the mis-calibration is a conservative one: the
1128 model-based standard error of the interaction is over-estimated by 6 to 10 percent
1129 and the realised type-I error is approximately 0.03 against a nominal 0.05.

1130 Second, the four-channel decomposition localises the discrepancy to the standard-
1131 error channel and excludes point-estimate bias, the degrees-of-freedom rule, and
1132 the positive-definiteness correction.

1133 Third, the working-covariance model ladder localises the cause to the corCAR1
1134 residual layer. A random intercept alone is well calibrated; adding the corCAR1
1135 residual removes the calibration.

1136 Fourth, the sample-size gradient shows the mis-calibration is asymptotic. It does
1137 not attenuate from $N = 35$ to $N = 280$, and so it is a property of the work-
1138 ing covariance and not a small-sample artefact amenable to a Kenward-Roger
1139 correction.

1140 Fifth, a CR2 bias-reduced cluster-robust standard error, clustered on participant
1141 and referred to a Satterthwaite distribution, restores the standard-error ratio to

approximately one and the type-I error to approximately 0.05, without altering the point estimate or the analysis model. At an alternative cell the same recalibration is a near-uniform level shift in power that preserves the comparative ranking of specifications.

Sixth, and in our view most consequential for routine practice, the staged protocol, comprising the four-channel decomposition, the model ladder, and the sample-size gradient, is a reusable instrument for any mixed-model interaction test whose calibration is in doubt. We recommend that interaction tests reported from mixed models with a non-trivial working covariance be accompanied by a calibration check of this kind, or by a cluster-robust standard error that is agnostic to the working covariance.

7 Reproducibility

All diagnostic outputs are generated by six named scripts and read inline; the manuscript is fully reproducible from the repository root.

- Six scripts under `analysis/scripts/carryover-sensitivity/` cover the decomposition, model ladder, sample-size gradient, null and alternative CR2 verifications, and GEE check.
- Each script writes a replicate-level `.rds` file; this manuscript reads those files and regenerates every table and figure inline.
- The examination plan that preceded the manuscript is retained alongside it as `type1-examination-plan.md`.

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The diagnosis is driven by six scripts under `analysis/scripts/carryover-sensitivity/`: `diagnose-null-type1.R` for the four-channel decomposition, `diagnose-null-type1-stages46.R` for the model ladder and the sample-size gradient, `diagnose-null-type1-cr2.R` for the cluster-robust verification at the null, `diagnose-alt-cell-cr2.R` for the alternative-cell check, and `diagnose-null-type1-gee.R` for the GEE marginal-model check, with `check-null-architecture-equivalence.R` establishing the architecture-invariance of the null. Each writes a replicate-level `.rds` file into the `output/` subdirectory, and this manuscript reads those files and regenerates every table and figure inline. The cluster-robust standard errors use the `clubSandwich` package. The examination plan that preceded the manuscript is retained alongside it as `type1-examination-plan.md`.

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